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#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

#define MAX 10   // Maximum parking slots
#define PASSWORD "admin123"  // Admin password

struct Vehicle {
    char num[20];
    char type[10];
    time_t entryTime;  // stores time in seconds
};

int count = 0;  // Current vehicles
struct Vehicle park[MAX];

void login();
void vehicleEntry();
void vehicleExit();
void showStatus();
void saveToFile();
double calculateFee(time_t entry, time_t exit);

int main() {
    int choice;
    login(); // login first

    while (1) {
        printf("\n========== VEHICLE PARKING SYSTEM ==========\n");
        printf("1. Park Vehicle\n");
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        printf("2. Exit Vehicle\n");
        printf("3. Show Parking Status\n");
        printf("4. Save & Exit\n");
        printf("-----------------------------------------\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                vehicleEntry();
                break;
            case 2:
                vehicleExit();
                break;
            case 3:
                showStatus();
                break;
            case 4:
                saveToFile();
                printf("\nData saved. Thank you!\n");
                exit(0);
            default:
                printf("\nInvalid Choice! Try Again.\n");
        }
    }
    return 0;
}

//  Admin login
void login() {
    char pass[20];
```

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    printf("\n===== ADMIN LOGIN =====\n");
    tryAgain:
    printf("Enter Password: ");
    scanf("%s", pass);

    if (strcmp(pass, PASSWORD) != 0) {
        printf("Wrong Password! Try Again.\n");
        goto tryAgain;
    }
    printf("Login Successful!\n");
}

//  Add a vehicle
void vehicleEntry() {
    if (count >= MAX) {
        printf("\nParking Full! No Slot Available.\n");
        return;
    }

    printf("\n--- Vehicle Entry ---\n");
    printf("Enter Vehicle Number: ");
    scanf("%s", park[count].num);
    printf("Enter Vehicle Type (Car/Bike/Bus): ");
    scanf("%s", park[count].type);

    park[count].entryTime = time(NULL);  // store entry time
    count++;

    printf("\nVehicle Parked Successfully!\n");
}
```

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//  Remove a vehicle + show fee
void vehicleExit() {
    if (count == 0) {
        printf("\nNo Vehicles Parked.\n");
        return;
    }

    char num[20];
    printf("\n--- Vehicle Exit ---\n");
    printf("Enter Vehicle Number: ");
    scanf("%s", num);

    for (int i = 0; i < count; i++) {
        if (strcmp(park[i].num, num) == 0) {
            time_t exitTime = time(NULL);
            double fee = calculateFee(park[i].entryTime, exitTime);

            printf("\nVehicle Found!\n");
            printf("Type: %s\n", park[i].type);
            printf("Parking Fee: ₹%.2f\n", fee);

            // Remove vehicle by shifting array
            for (int j = i; j < count - 1; j++)
                park[j] = park[j + 1];

            count--;
            printf("Vehicle Removed Successfully!\n");
            return;
        }
    }
    printf("\nVehicle Not Found!\n");
```

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}

//  Show all parked vehicles
void showStatus() {
    if (count == 0) {
        printf("\nNo Vehicles Parked.\n");
        return;
    }

    printf("\n========== CURRENTLY PARKED VEHICLES ==========\n");
    printf("No.\tNumber\t\tType\t\tEntry Time\n");
    printf("-----------------------------------------------\n");

    for (int i = 0; i < count; i++) {
        char *timeStr = ctime(&park[i].entryTime);
        timeStr[strlen(timeStr)-1] = '\0'; // remove newline

        printf("%d\t%s\t\t%s\t\t%s\n", i+1, park[i].num, park[i].type, timeStr);
    }
}

//  Save data to file
void saveToFile() {
    FILE *fp = fopen("parking_data.txt", "w");
    if (fp == NULL) {
        printf("\nError Saving File!\n");
        return;
    }

    for (int i = 0; i < count; i++) {
        fprintf(fp, "%s %s %lld\n", park[i].num, park[i].type, (long long)park[i].entryTime);
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```
    }
    fclose(fp);
}

//  Fee Calculation (per hour basis)
double calculateFee(time_t entry, time_t exit) {
    double diff = difftime(exit, entry); // seconds
    double hours = diff / 3600.0;

    if (hours < 1) hours = 1; // minimum charge

    return hours * 20; // ₹20 per hour
}
```